

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Ratios, rates, proportional relationships, and units

04

10 answers · Rationale-rich · Teach from doc

Q1**A****A. 18**

Choice A is correct. Dividing the number of revolutions by the number of minutes gives the number of revolutions the turbine completes per minute. It's given that the wind turbine completes 900 revolutions in 50 minutes. Therefore, at this rate, this turbine completes $\frac{900}{50}$, or 18, revolutions per minute.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D****D. 192**

Choice D is correct. Since 1 yard is equal to 3 feet, 64 yards is equal to $64 \text{ yards} \frac{3 \text{ feet}}{1 \text{ yard}}$, or 192 feet. It follows that 64 yards per second is equivalent to 192 feet per second. Therefore, the object's speed is 192 feet per second.

Choice A is incorrect. A speed of 61 feet per second is equivalent to $\frac{61}{3}$, not 64, yards per second.

Choice B is incorrect. A speed of 67 feet per second is equivalent to $\frac{67}{3}$, not 64, yards per second.

Choice C is incorrect. A speed of 94 feet per second is equivalent to $\frac{94}{3}$, not 64, yards per second.

Q3**39000**

The correct answer is 39,000. It's given that the giant armadillo has a mass of 39 kilograms. Since 1 kilogram is equal to 1,000 grams, 39 kilograms is equal to $39 \text{ kilograms} \times \frac{1,000 \text{ grams}}{1 \text{ kilogram}}$, or 39,000 grams. Therefore, the giant armadillo's mass, in grams, is 39,000.

Q4**102**

The correct answer is 102. It's given that 1 yard is equivalent to 3 feet. Therefore, 34 yards is equivalent to $34 \text{ yards} \times \frac{3 \text{ feet}}{1 \text{ yard}}$, or 102 feet.

Q5**31**

The correct answer is 31. It's given that 1 yard is equal to 36 inches. Therefore, 1,116 inches is equivalent to $1,116 \text{ inches} \times \frac{1 \text{ yard}}{36 \text{ inches}}$, or 31 yards.

Q6**A****A. 17**

Choice A is correct. It's given that the ratio of the length of line segment XY to the length of line segment ZV is 6 to 1, which means $\frac{XY}{ZV} = \frac{6}{1}$. It's given that the length of line segment XY is 102 inches. If the length, in inches, of line segment ZV is represented by l , the value of l can be calculated by solving the equation $\frac{102}{l} = \frac{6}{1}$, or $\frac{102}{l} = 6$. Multiplying each side of this equation by l yields $102 = 6l$. Dividing each side of this equation by 6 yields $17 = l$. Therefore, the length of line segment ZV is 17 inches.

Choice B is incorrect. This is the length, in inches, of line segment ZV if the length of line segment XY is 576, not 102, inches.

Choice C is incorrect. This is the length, in inches, of line segment XY , not line segment ZV .

Choice D is incorrect. This is the length, in inches, of line segment ZV if the ratio of the length of line segment XY to the length of line segment ZV is 1 to 6, not 6 to 1.

Q7**A****A. 18**

Choice A is correct. It's given that there are 2,358 raccoons in a 131-square-mile area. The estimated population density, in raccoons per square mile, is the estimated number of raccoons divided by the number of square miles. Therefore, the estimated population density of this area is $\frac{2,358 \text{ raccoons}}{131 \text{ square miles}}$, or 18 raccoons per square mile.

Choice B is incorrect. This is the number of square miles in the area, not the estimated number of raccoons per square mile in this area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**A****A.** 1 to 4

Choice A is correct. It's given that Shaquan has 7 red cards and 28 blue cards. Therefore, the ratio of red cards to blue cards that Shaquan has is 7 to 28. This ratio can be reduced by dividing both parts of the ratio by 7, which yields the ratio 1 to 4.

Choice B is incorrect. This is the ratio of blue cards to red cards that Shaquan has. Choice C is incorrect and may result from a calculation error when reducing the ratio. Choice D is incorrect. This may result from finding the ratio of blue cards to red cards, or 28 to 7, and then making a calculation error when reducing the ratio.

Q9**C****C.** 24

Choice C is correct. It's given that the cherry pitting machine pits 12 pounds of cherries in 3 minutes. This rate can be written as $\frac{12 \text{ pounds of cherries}}{3 \text{ minutes}}$. If the number of minutes it takes the machine to pit 96 pounds of cherries is represented by x , the value of x can be calculated by solving the equation $\frac{12 \text{ pounds of cherries}}{3 \text{ minutes}} = \frac{96 \text{ pounds of cherries}}{x \text{ minutes}}$, which can be rewritten as $\frac{12}{3} = \frac{96}{x}$, or $4 = \frac{96}{x}$. Multiplying each side of this equation by x yields $4x = 96$. Dividing each side of this equation by 4 yields $x = 24$. Therefore, it takes the machine 24 minutes to pit 96 pounds of cherries.

Choice A is incorrect. This is the number of minutes it takes the machine to pit 32, not 96, pounds of cherries.

Choice B is incorrect. This is the number of minutes it takes the machine to pit 60, not 96, pounds of cherries.

Choice D is incorrect. This is the number of minutes it takes the machine to pit 144, not 96, pounds of cherries.

Q10

180

The correct answer is 180. It's given that a mechanical device produces items at a constant rate of 60 items per hour. This rate can be written as $\frac{60 \text{ items}}{1 \text{ hour}}$. Let x represent the number of items the mechanical device will produce in 3 hours at the given rate. It follows that $\frac{60 \text{ items}}{1 \text{ hour}} = \frac{x \text{ items}}{3 \text{ hours}}$, which can be written as $\frac{60}{1} = \frac{x}{3}$, or $60 = \frac{x}{3}$. Multiplying each side of this equation by 3 yields $180 = x$. Therefore, at the given rate, the mechanical device will produce 180 items in 3 hours.

Alternate approach: It's given that a mechanical device produces items at a constant rate of 60 items per hour. At this rate, the mechanical device will produce $\frac{60 \text{ items}}{1 \text{ hour}} \cdot 3 \text{ hours}$, or 180 items in 3 hours.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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